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ABOUT THE JOURNAL

Aims and Scope

Cell Discovery is a fully open access international journal that publishes results of significance and originality in all areas of molecular & cell biology. *Cell Discovery* is established in 2015 as a sister journal of [Cell Research](#), a high profile international journal.

Cell Discovery, published by Springer Nature in partnership with the Center for Excellence in Molecular Cell Science, Chinese Academy of Sciences (CAS), aims to provide an open access platform for scientists to publish their outstanding original works. The basic bar of acceptance is comparable to prestigious society journals in the respective field of the work.

The journal has an international authorship and a broad scope in basic life sciences, and is a valued member of

Springer Nature's highly respected family of Molecular Cell Biology journals.

Journal Details

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ARTICLE TYPES

Article Description	Abstract	Word Limit	Tables/ Figures	References guideline
Article A complete, comprehensive report of original research. An Article usually has a fairly complex narrative that is based on multiple techniques and/or approaches.	Unstructured abstract; max word limit: 250	6,000 – 8,000 words excluding abstract, references, figures and tables	Max of 8	Max of 100. Please use as current as possible.
Correspondence Correspondence provides rapid and concise report of a novel finding that is brief in nature but is of general interest to the field. There should be no more than 10 authors.	None	1,200 words excluding, references, figures and tables	Max of 1	Max of 10,
Editorial The editors invite editorials to discuss a topical issue or a paper published in the journal and set the problems addressed by the paper in the wider context of the field. These are usually commissioned, but unsolicited editorial submissions are considered for publication.	None	1,000 words excluding references	Max of 1	Max of 10
Review Article A comprehensive synthesis and/or analysis of specific topics. A short Introduction giving the rationale for the review should be followed by	Unstructured abstract; max word limit: 250	15,000 words max excluding abstract, references,	Max of 8	Max of 250

sections with appropriate subheadings, followed by a conclusions section at the end. The standard footer headings (Acknowledgements, Contributions, Competing Interests, Funding) are required. All invited reviews will undergo peer review prior to acceptance.		figures and tables		
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Word limit

Word limits are provided for guidance only. The Editors will consider submissions that exceed the recommended limit, subject to feedback received during peer review.

PREPARATION OF ARTICLES

Please note that Articles must contain the following components in the order stated. Please see below for further details.

- Title page
- Abstract
- Introduction
- Results
- Discussion
- Materials and methods
- Acknowledgements
- Conflict of interests
- Contributions
- References
- Figure legends
- Tables
- Figures

Article Requirements

Cover letter

Each manuscript must be accompanied by a cover letter including statements that:

- Highlight of the current manuscript should be limited to no more than 2 short sentences;
- All authors agree with the submission;
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- Conflict of Interest Statement

Organization of manuscript

All textual content should be provided in a single file, prepared using either Microsoft Word or TeX/LaTeX; figures should be provided in individual files. The manuscript text file should include the following parts, in order: a title page with author affiliations and contact information (the corresponding author should be identified with an asterisk); the sections required for each content type (see information for different content types) then References, Acknowledgements (optional), Author

Contributions (Articles only), Competing Financial Interests statement, Figure Legends and Tables. Footnotes to the text are not allowed and any such material should be incorporated into the text as parenthetical matter.

(i) Title page

The title page should include a succinct title (less than 200 characters); a concise running title (which should normally not exceed 50 characters); the full names of all authors including their given names; the affiliations (including city, state, country and zip code) of all authors; the official email addresses of all authors, and the full contact details of the corresponding author (including telephone and fax numbers).

(ii) Abstract

A brief abstract (maximum 250 words) should state the purpose, basic procedures, main findings and principal conclusions of the study. The abstract should not contain abbreviations or references and should not be structured.

(iii) Introduction

The Introduction should summarize the rationale for the study and outline pertinent background material. The Introduction should not contain either results or conclusions.

(iv) Results

The Results should be presented in a logical sequence in the text, tables and figures; repetitive presentation of the same data in different forms should be avoided. The Results should not include material appropriate to the Discussion.

(v) Discussion

The Discussion should not reiterate Results, but rather should consider them in relation to any hypotheses advanced in the Introduction. This may include an evaluation of methodology and the relationship of new information to the existing body of knowledge in that field.

(vi) Materials and methods

Materials and Methods should be described in sufficient detail to allow the experimental work to be reproduced in another laboratory, and to leave the reader in no doubt as to how the results were derived.

Data Availability Statement and Policy: An inherent principle of publication is that others should be able to

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It also means that a Data Availability Statement ([see here for more details](#)) must be included as part of your manuscript.

Sequences, Structures and "Omics"

Papers reporting protein or DNA sequences and molecular structures will not be accepted without an accession number to [Genbank](#), [EMBL](#), [DDBJ](#), [Uniprot](#), [ProteinDataBank](#), or other publicly available database in general use in the field that gives free access to researchers from the date of publication.

Authors of papers describing structures of biological macromolecules must provide experimental data upon the request of Editor if they are not already freely accessible in a publicly available database such as [ProteinDataBank](#), [Biological Magnetic Resonance Databank](#), or [Nucleic Acid Database](#).

(vii) Acknowledgments

Authors should acknowledge the source of financial grants and other funding, and declare any industrial links or affiliations. The contribution of colleagues or institutions should also be acknowledged. Personal thanks and thanks to anonymous reviewers should not be included.

(viii) Conflict of interests

A conflict of interest statement must be included for each contributing author. Please see the Conflict of Interest guidelines in the Editorial Policies section for more information and for guidelines on what constitutes a conflict of interest.

(ix) References

References: All necessary references should be included in order to credit previous work directly relevant to the article. References should follow the *Nature* style available in most reference management software. In the text they should appear as superscript numbers starting at 1 and at the end of the paper they should be listed (double-spaced) in numerical order corresponding to the order of citation in the text. Where a reference is to appear next to a number in the text, for example, following an equation, chemical formula or biological acronym, citations should be written as (ref. X) and not as superscript. Example: "detectable levels of endogenous Bcl-2 (ref. 3), as confirmed by western blot."

All authors should be listed for papers with up to five authors; for papers with more than five authors, the first only should be listed, followed by *et al.* Abbreviations for titles of medical periodicals should conform to those used in the latest edition of Index Medicus. The first and last page numbers for each reference should be provided. Abstracts

must be identified as such. Papers in press and preprints hosted on a recognized server may be included in the list of references.

Personal communications must be allocated a number and included in the list of references in the usual way or simply referred to in the text; the authors may choose which method to use. In either case authors must obtain permission from the individual concerned to quote his/her unpublished work.

Examples:

Journal article, up to five authors:

Belkaid, Y. & Rouse, B. T. Natural regulatory T cells in infectious disease. *Nat. Immunol.* **6**, 353–360 (2005).

Journal article, e-pub ahead of print:

Bonin, M. *et al.* F-ara-A pharmacokinetics during reduced-intensity conditioning therapy with fludarabine and busulfan. *Bone Marrow Transplant.* <http://dx.doi.org/10.1038/sj.bmt.1705565> (2007).

Journal article, in press:

Gallardo, R. L., Juneja, H. S. & Gardner, F. H. Normal human marrow stromal cells induce clonal growth of human malignant T-lymphoblasts. *Int. J. Cell Cloning* (in the press).

Complete book:

Atkinson, K. *et al.* (eds) *Clinical Bone Marrow and Blood Stem Cell Transplantation* (Cambridge Univ. Press, 2004).

Chapter in book:

Harley, N. H. & Vivian, L. in *Mechanisms of Disease* 4th edn, Vol. 2 (eds Sodeman, W. A. & Smith, A.) Ch. 3 (Saunders, 1974).

Abstract:

Feig, S. A. *et al.* Bone marrow transplantation for neuroblastoma. *Exp. Hematol.* **13**, abstr. 102 (1985).

Preprint:

Starrfelt, J. & Liow, L.H. How many dinosaur species were there? Fossil bias and true richness estimated using a Poisson sampling model (TRiPS). Preprint at <http://biorxiv.org/content/early/2015/12/04/025940> (2015).

Research dataset:

Hao, Z., AghaKouchak, A., Nakhjiri, N. & Farahmand, A. Global Integrated Drought Monitoring and Prediction System (GIDMaPS) Data sets. [figshare](http://dx.doi.org/10.6084/m9.figshare.853801). <http://dx.doi.org/10.6084/m9.figshare.853801> (2014).

(x) Figures

Figures and images should be labelled sequentially, numbered and cited in the text. Production-quality figures are not required at initial submission, but to avoid potential substantial revisions at later stages you may wish to note some of the guidelines below even at the initial submission stage.

It is recommended that you convert all your figures to JPEG before generating PDFs or uploading individual files. This will reduce the file sizes and the amount of time it takes the files to upload to our submission site and will also give you a closer approximation to the way your figures will appear on our site. If you choose to submit your files in PowerPoint format, please do not make a JPEG of these within PowerPoint. The conversion is more successful when a raw PowerPoint file is submitted.

General Figure Guidelines

Use distinct colours with comparable visibility and consider colour-blind individuals by avoiding the use of red and green for contrast. Recoloring primary data, such as fluorescence images, to colour-safe combinations such as green and magenta, turquoise and red, yellow and blue or other accessible colour palettes is strongly encouraged. Use of the rainbow colour scale should be avoided. Use solid colour for filling objects and avoid hatch patterns. Avoid background shading. Figures divided into parts should be labelled with a lower-case, boldface 'a', 'b', etc. in the top left-hand corner. Labelling of axes, keys and so on should be in 'sentence case' (first word capitalized only) with no full stop. Units must have a space between the number and the unit, and follow the nomenclature common to your field. Commas should be used to separate thousands. Unusual units or abbreviations should be spelled out in full, or defined in the legend.

Final Figure Submission Guidelines

Should your manuscript be accepted, you will receive more extensive instructions for final submission of display items. However, a summary of our guidelines for final figure preparation are included here.

- Each figure should be saved in a separate file. Figures including multiple parts (e.g. Fig.1a, 1b, 1c) should be saved in a single file (e.g. Figure1a-c). The figure number should be placed above each figure. Figure legends should be inserted in the article's text file.
- Images should be saved in RGB color mode at 300 dpi or higher resolution.
- Use the same typeface (Arial, Helvetica or Times New Roman) for all figures. Use symbol font for Greek letters.
- We prefer vector files with editable layers. Acceptable formats are: .ai, .eps, .pdf, .ps, .svg for fully editable vector-based art; layered .psd or .tiff for editable layered art; .psd, .tif, .jpeg or .png for bitmap images; .ppt if fully editable and without styling effects; ChemDraw (.cdx) for chemical structures.
- Figures are best prepared at the size you would expect them to appear in print. At this size, the optimum font size is 8pt and no lines should be thinner than 0.25 pt (0.09 mm).

Display items that contain chemical structures should be produced using ChemDraw or a similar program. Authors using ChemDraw should use our ChemDraw Template and submit the final files at 100% as .cdx files. All chemical compounds must be assigned a bold, Arabic numeral in the

order in which the compounds are presented in the manuscript text.

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Tables should be labelled sequentially as Table 1, Table 2, etc. Each table should be numbered, titled and cited in the text. Reference to table footnotes should be made by using Arabic numerals. Tables should not duplicate the content of the text. They should consist of at least two columns, and each column should have a heading. Authors should ensure that the data in the tables are consistent with those cited in the relevant places in the text, totals add up correctly, and percentages have been calculated correctly. Unlike figures or images, tables may be embedded into the main manuscript file if necessary, or supplied as separate electronic files.

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Artwork Guidelines

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Color figures must be supplied in the following format. For Single Images:

Width	500 pixels (authors should select "constrain proportions", or equivalent instructions, to allow the application to set the correct height automatically.)
Resolution	125 dpi (dots per inch)
Format	JPEG for photographs GIF for line drawings or charts
Filenaming	Please save image with .jpg or .gif extension to ensure it can be read by all platforms and graphics packages.

For Multi-part Images:

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Filenaming	Please save image with .jpg or .gif extension to ensure it can be read by all platforms and graphics packages.

Black and white images

- Image resolution of at least 300 dpi at publication size
- Images should be scanned at a minimum of 300 dpi
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- Contrast and brightness can be adjusted, but this must be uniform across the entire image, and must not result in the loss of any feature, band or spot. The background should still be visible
- If lanes are removed, and once separate parts of an image are joined together, a black, white or grey line should indicate clearly where the image was cut
- If black borders are drawn around the image, the lines should correspond to all edges where the image was cut
- Protein molecular weights or DNA fragment sizes should be indicated for all figure panels showing gel electrophoresis

Graphs, Histograms and Statistics

- Error bars must be described in the figure legend
- Axes on graphs should extend to zero, except for log axes
- Statistical analyses (including error bars and p values) should only be shown for independently repeated experiments, and must not be shown for replicates of a single experiment
- The number of times an experiment was repeated (N) must be stated in the legend

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1. Do not make rules thinner than 1 pt (0.36 mm)
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3. Color should be distinct when being used as an identifying tool
4. Use SI units throughout
5. Commas should be used to separate thousands
6. Abbreviations should be preceded by the words for which they stand in the first instance of use and should not be used for terms used fewer than 4 times
7. Text should be double spacing with a wide margin
8. Use a common word-processing package (such as Microsoft Word) for the text. Embed tables converted into images at the end of the Word document, or as a separate file in whichever program you used to generate them

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11. All pages and lines are to be numbered. To add page numbers in MS Word, go to Insert then Page Numbers. To add line numbers go to File, Page Setup, then click the Layout tab. In the Apply to box, select Whole document, click Line Numbers then select the Add line numbering check box, followed by Continuous.

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File formats for manuscript files, figures and tables that are acceptable for our electronic manuscript submission process are given on the online forms. Further advice on file types is also available from the [Tips](#) webpage. Please follow the artwork guidelines above for submitting figures, and use a common word-processing package (such as Microsoft Word) for the text. Either embed tables converted into images at the end of your Word document, or as a separate file in whichever program you used to generate them. If you submit raw data, this can be done in Excel, or tab/comma delimited format.

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Supplementary information is peer-reviewed material directly relevant to the conclusions of an article that cannot be included in the printed version owing to space or format constraints. It is posted on the journal's website and linked to the article when the article is published and may consist of data files, graphics, movies or extensive tables.

The article must be complete and self-explanatory without the supplementary information. Supplementary information enhances a reader's understanding of the manuscript but is not essential to that understanding.

Supplementary information must be supplied to the Editorial Office in its final form for peer review. On acceptance the final version of the peer reviewed supplementary information should be submitted with the accepted manuscript.

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Authors should ensure that supplementary information is supplied in its FINAL format because it is not subedited and will appear online exactly as originally submitted. It can neither be altered, nor added to, after the paper has been accepted for publication.

Please supply the supplementary information via eJP, the electronic manuscript submission and tracking system, in an acceptable file format (see below).

Authors should:

- Include a text summary (no more than 50 words) to describe the contents of each file.
- Identify the types of files (file formats) submitted.
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Accepted file formats

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- MPEG movie files (.mpg)
- JPEG image files (.jpg)
- Sound files (.wav)
- Plain ASCII text (.txt)
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- MS Word documents (.doc)
- Postscript files (.ps)
- MS Excel spreadsheet documents (.xls)
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File sizes must be as small as possible, so that they can be downloaded quickly. Images should not exceed 640 x 480 pixels (approximately 23 x 17 cm at 72dpi), and we would recommend 480 x 360 pixels as the maximum frame size for movies. We also recommend a frame rate of 15 frames per second. If applicable to the presentation of the supplementary information, use a 256-color palette. Please

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- (1) A rebuttal letter, indicating point-by-point how you have addressed the comments raised by the reviewers. If you disagree with any of the points raised, please provide adequate justification in your letter.
- (2) A marked-up version of the manuscript that highlights changes made in response to the reviewers' comments in order to aid the Editors and reviewers.
- (3) A 'clean' (non-highlighted) version of the manuscript.

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AND to have approved the submitted version (and any substantially modified version that involves the author's contribution to the study);

AND to have agreed both to be personally accountable for the author's own contributions and to ensure that questions related to the accuracy or integrity of any part of the work, even ones in which the author was not personally involved, are appropriately investigated, resolved, and the resolution documented in the literature.

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In the interests of transparency and to help readers form their own judgments of potential bias, authors must declare whether or not there are any competing financial interests in relation to the work described. The corresponding author is responsible for submitting a competing financial interests statement on behalf of all authors of the paper. This statement must be included on the title page as well as within the article before the References section listed under 'Competing Interests'.

In cases where the authors declare a competing financial interest, a statement to that effect is published as part of the article. If no such conflict exists, the statement will simply read that the authors have nothing to disclose.

For the purposes of this statement, competing interests are defined as those of a financial nature that, through their potential influence on behaviour or content, or from perception of such potential influences, could undermine the objectivity, integrity or perceived value of a publication. They can include any of the following:

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- *Patents*: Holding, or currently applying for, patents, relating to the content of a manuscript; receiving reimbursement, fees, funding, or salary from an organization that holds or has applied for patents relating to the content of the manuscript.

It is difficult to specify a threshold at which a financial interest becomes significant, but note that many US universities require faculty members to disclose interests exceeding \$10,000 or 5% equity in a company. Any such figure is arbitrary, so we offer as one possible practical alternative guideline: "Declare all interests that could embarrass you were they to become publicly known after your work was published." We do not consider diversified mutual funds or investment trusts to constitute a competing financial interest.

The statement included in the submission must contain an explicit and unambiguous description of any potential competing interests, or lack thereof, for any of the authors as it relates to the subject of the report. Examples include:

- **Competing Interests**

The authors declare no competing financial interests

- **Competing Interests**

Dr Caron's work has been funded by the NIH. He has received compensation as a member of the scientific advisory board of Acadia Pharmaceutical and owns stock in the company. He also has consulted for Lundbeck and received compensation. Dr Rothman and Dr Jensen declare no potential competing interests.

Neither the precise amount received from each entity nor the aggregate income from these sources needs to be provided.

Non-financial interests that authors may like to disclose include:

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- an academic link or rivalry with someone whose interests may be affected by publication of the article,
- membership in a political party or special interest group whose interests may be affected by publication of the article, or
- a deep personal or religious conviction that may have affected what the author wrote and that readers should be aware of when reading the article.

Studies involving animals and other human subjects

Research involving human subjects, human material, or human data must have been performed in accordance with the Declaration of Helsinki and must have been approved by an appropriate ethics committee. A statement detailing this, including the name of the ethics committee and the reference number where appropriate, along with a statement confirming that informed consent was obtained from all subjects, must appear in all manuscripts reporting such research.

For primary research manuscripts reporting experiments on live vertebrates and/or higher invertebrates, the corresponding author must confirm that all experiments

were performed in accordance with relevant guidelines and regulations. The manuscript must include in the Supplementary Information (methods) section (or, if brief, within of the print/online article at an appropriate place), a statement identifying the institutional and/or licensing committee approving the experiments, including any relevant details regarding animal welfare, patient anonymity, drug side effects and informed consent. Sex and other characteristics of animals that may influence results must be described. Details of housing and husbandry must be included where they are likely to influence experimental results. *Cell Discovery* recommends following the [ARRIVE reporting guidelines](#) when documenting animal studies.

Clinical Trials

All clinical trials must be registered in a public registry prior to submission and the trial registry number must be included in the manuscript and provided on submission. The journal follows the trials registration policy of the ICMJE (www.icmje.org) and considers only trials that have been appropriately registered before submission, regardless of when the trial closed to enrolment. Acceptable registries must meet the following ICMJE requirements:

- be publicly available, searchable, and open to all prospective registrants
- have a validation mechanism for registration data
- be managed by a not-for-profit organization

Examples of registries that meet these criteria include:

1. ClinicalTrials.gov the registry sponsored by the United States National Library of Medicine
2. the [International Standard Randomized Controlled Trial Number Registry](#)
3. the [Cochrane Renal Group Registry](#)
4. the [European Clinical Trials Database](#)

Randomised Controlled Trials (RCTs) must adhere to the CONSORT statement, (CONSolidated Standards Of Reporting Trials) and submissions must be accompanied by a completed CONSORT checklist (uploaded as a related manuscript file). Further information can be found at www.consort-statement.org.

Informed Consent

When publishing identifiable images from human research participants, authors must include a statement attesting that they have obtained informed consent for publication of the images. If the participant is deceased, consent must be sought from the next of kin of the participant. All reasonable measures must be taken to protect patient anonymity. Black bars over the eyes are not acceptable means of anonymization. In certain cases, the journal may insist upon obtaining evidence of informed consent from authors. Images without appropriate consent will be removed from publication.

Cell Line Authentication

If human cell lines are used, authors are strongly encouraged to include the following information in their manuscript:

- the source of the cell line, including when and from where it was obtained,
- whether the cell line has recently been authenticated and by what method, and
- whether the cell line has recently been tested for mycoplasma contamination.

Further information is available from [the International Cell Line Authentication Committee](#) (ICLAC). We recommend that authors check the [NCBI database](#) for misidentification and contamination of human cell lines.

Biosecurity Policy

The Editor may seek advice about submitted papers not only from technical reviewers but also on any aspect of a paper that raises concerns. These may include, for example, ethical issues or issues of data or materials access. Occasionally, concerns may also relate to the implications to society of publishing a paper, including threats to security. In such circumstances, advice will usually be sought simultaneously with the technical peer-review process. As in all publishing decisions, the ultimate decision whether to publish is the responsibility of the editor.

Reproducibility

Cell Discovery requires authors of papers that are sent for external review to include in their manuscripts relevant details about several elements of experimental and analytical design. This initiative aims to improve the transparency of reporting and the reproducibility of published results, focusing on [elements of methodological information](#) that are frequently poorly reported.

Misconduct

Springer Nature takes seriously all allegations of potential misconduct. As a member of the [Committee on Publication Ethics](#) (COPE), *Cell Discovery* will follow the COPE guidelines outlining how to deal with cases of suspected misconduct. As part of the investigation, the journal may opt to do one or more of the following:

- suspend review or publication of a paper until the issue has been investigated and resolved;
- request additional information from the author, including original data or images or ethics committee or IRB approval;
- make inquiries of other titles believed to be affected;
- forward concerns to the author's employer or person responsible for research governance at the author's institution;
- refer the matter to other authorities or regulatory bodies (for example, the Office of Research Integrity in the US or the General Medical Council in the UK); or
- submit the case to COPE in an anonymized form for additional guidance on resolution.

Please note that, in keeping with the journal's policy of the confidentiality of peer review, if sharing of information

with third parties is necessary, disclosure will be made to only those Editors who the Editor believes may have information that is pertinent to the case, and the amount of information will be limited to the minimum required.

Duplicate Publication

Papers must be original and not published or submitted for publication elsewhere. This rule also applies to non-English language publications.

Springer Nature allows and encourages prior publication on recognized community preprint servers for review by other scientists before formal submission to a journal. The details of the preprint server concerned and any accession numbers should be included in the cover letter accompanying manuscript submission. This policy does not extend to preprints available to the media or that are otherwise publicized outside the scientific community before or during the submission and consideration process.

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Plagiarism

Plagiarism is when an author attempts to pass off someone else's work as his or her own. Duplicate publication, sometimes called self-plagiarism, occurs when an author reuses substantial parts of his or her own published work without providing the appropriate references. This can range from getting an identical paper published in multiple journals, to 'salami-slicing', where authors add small amounts of new data to a previous paper.

Plagiarism can be said to have clearly occurred when large chunks of text have been cut-and-pasted. Minor plagiarism without dishonest intent is relatively frequent, for example, when an author reuses parts of an introduction from an earlier paper. Journal editors judge any case of which they become aware (either by their own knowledge of and reading about the literature, or when alerted by referees) on its own merits.

Springer Nature is a member of Similarity Check (formerly CrossCheck), a multi-publisher initiative used to screen published and submitted content for originality. *Cell Discovery* uses Similarity Check to detect instances of overlapping and similar text in submitted manuscripts. To find out more about visit the [Similarity Check](#) website.

If a case of plagiarism comes to light after a paper is published, the Journal will conduct a preliminary investigation, utilising the guidelines of the [Committee on Publication Ethics](#). If plagiarism is proven, the Journal will contact the author's institute and funding agencies as appropriate. The paper containing the plagiarism may also be formally retracted or subject to correction.

Data Fabrication & Falsification

Falsification is the practice of altering research data with the intention of giving a false impression. This includes, but is not limited to, manipulating images, removing outliers or “inconvenient” results, or changing, adding or omitting data points. Fabrication is the practice of inventing data or results and recording and/or reporting them in the research record. Data falsification and fabrication call into question the integrity and credibility of data and the data record, and as such, they are among the most serious issues in scientific ethics.

Some manipulation of images is allowed to improve them for readability. Proper technical manipulation includes adjusting the contrast and/or brightness or colour balance if it is applied to the complete digital image (not parts of the image). The author should notify the Editor in the cover letter of any technical manipulation. Improper technical manipulation refers to obscuring, enhancing, deleting and/or introducing new elements into an image. See Image Integrity & Standards below for more details.

Image Integrity and Standards

Images submitted with a manuscript for review should be minimally processed (for instance, to add arrows to a micrograph). Authors should retain their unprocessed data and metadata files, as editors may request them to aid in manuscript evaluation. If unprocessed data is unavailable, manuscript evaluation may be stalled until the issue is resolved.

A certain degree of image processing is acceptable for publication, but the final image must correctly represent the original data and conform to community standards. The guidelines below will aid in accurate data presentation at the image processing level:

- Authors should list all image acquisition tools and image processing software packages used. Authors should document key image-gathering settings and processing manipulations in the Methods section.
- Images gathered at different times or from different locations should not be combined into a single image, unless it is stated that the resultant image is a product of time-averaged data or a time-lapse sequence. If juxtaposing images is essential, the borders should be clearly demarcated in the figure and described in the legend.
- Touch-up tools, such as cloning and healing tools in Photoshop, or any feature that deliberately obscures manipulations, is to be avoided.
- Processing (such as changing brightness and contrast) is appropriate only when it is applied equally across the entire image and is applied equally to controls. Contrast should not be adjusted so that data disappear. Excessive manipulations, such as processing to emphasize one region in the image at the expense of others (for example, through the use of a biased choice of threshold settings), is inappropriate, as is

emphasizing experimental data relative to the control.

For **gels and blots**, positive and negative controls, as well as molecular size markers, should be included on each gel and blot – either in the main figure or an expanded data supplementary figure. The display of cropped gels and blots in the main paper is encouraged if it improves the clarity and conciseness of the presentation. In such cases, the cropping must be mentioned in the figure legend.

- Quantitative comparisons between samples on different gels/blots are discouraged; if this is unavoidable, the figure legend must state that the samples derive from the same experiment and that gels/blots were processed in parallel. Vertically sliced gels that juxtapose lanes that were not contiguous in the experiment must have a clear separation or a black line delineating the boundary between the gels. Loading controls (e.g. GAPDH, actin) must be run on the same blot. Sample processing controls run on different gels must be identified as such, and distinctly from loading controls
- Cropped gels in the paper must retain important bands.
- Cropped blots in the body of the paper should retain at least six band widths above and below the band.
- High-contrast gels and blots are discouraged, as overexposure may mask additional bands. Authors should strive for exposures with gray backgrounds. Immunoblots should be surrounded by a black line to indicate the borders of the blot, if the background is faint.
- For quantitative comparisons, appropriate reagents, controls and imaging methods with linear signal ranges should be used.

Microscopy adjustments should be applied to the entire image. Threshold manipulation, expansion or contraction of signal ranges and the altering of high signals should be avoided. If ‘pseudo-colouring’ and nonlinear adjustment (for example ‘gamma changes’) are used, this must be disclosed. Adjustments of individual colour channels are sometimes necessary on ‘merged’ images, but this should be noted in the figure legend.

We encourage inclusion of the following with the final revised version of the manuscript for publication:

- In the Methods section, specify the type of equipment (microscopes/objective lenses, cameras, detectors, filter model and batch number) and acquisition software used. Although we appreciate that there is some variation between instruments, equipment settings for critical measurements should also be listed.
- The display lookup table (LUT) and the quantitative map between the LUT and the bitmap should be provided, especially when rainbow pseudo-colour is used. It should be stated if the LUT is linear and covers the full range of the data.

- Processing software should be named and manipulations indicated (such as type of deconvolution, three-dimensional reconstructions, surface and volume rendering, 'gamma changes', filtering, thresholding and projection).
- Authors should state the measured resolution at which an image was acquired and any downstream processing or averaging that enhances the resolution of the image.

Correction and Retraction Process

Publishable amendments that affect the publication record and/or the scientific accuracy of published information will receive a DOI and be published in the journal. Five categories of amendments are relevant for peer-reviewed material – detailed below.

All five correction types are bi-directionally linked to the original published paper.

Erratum or Publisher Correction. Notification of an important error made by the journal that affects the publication record or the scientific integrity of the paper, or the reputation of the authors or of the journal.

Corrigendum or Author Correction. Notification of an important error made by the author(s) that affects the publication record or the scientific integrity of the paper, or the reputation of the authors or the journal.

Retraction. Notification of invalid results that affect the reliability of a previously published article. The original article is marked as retracted but remains available to readers, and the retraction statement notifying readers of the invalidity of the published paper is bi-directionally linked to the original published paper.

Addendum. Notification of additional information about a paper. Addenda are published when the editors decide that the addendum is crucial to the reader's understanding of a significant part of the published contribution. Addenda include Editorial Expression of Concern, which is an editorial statement alerting our readership to serious concerns with the published paper. Editorial Expressions of Concern are typically updated with another amendment once further information is available.

Editor's Note. An editor's note is a statement from editors notifying readers of issues related to the published paper. It is an online update made only to the HTML version of record of the published article. Editor's notes are typically updated with another amendment once further information is available.

All corrections requests should be submitted to the [editorial office](#).

Decisions about corrections are made by the Editor (sometimes with peer-reviewers' advice) and this sometimes involves author consultation. Requests to make corrections that do not affect the paper in a significant way or impair the reader's understanding of the contribution (a

spelling mistake or grammatical error, for example) are not considered.

In cases where co-authors disagree about a correction, the editors will take advice from independent peer-reviewers and impose the appropriate correction, noting the dissenting author(s) in the text of the published version.

If there is suspicion of misconduct, the journal will carry out an investigation following COPE guidelines. Following an investigation, if the allegation raises valid concerns, the author will be contacted and given an opportunity to address the issue. If misconduct is established beyond reasonable doubt, this may result in the Editor implementing one of the following measures:

- If the article is still under consideration, it may be rejected and returned to the author.
- If the article has already been published online, depending on the nature and severity of the infraction, either a correction will be published alongside the article or, in severe cases, complete retraction of the article will occur. The reason for the correction or retraction must be given.

In either case, the author's institution or funding agency may be informed.

Supplementary information for editors and reviewers

Any manuscripts under review or accepted for publication elsewhere should accompany the submission if they are relevant to its scientific assessment.

Authors should also provide upon submission any kind of supplementary material that will aid the review process.

Peer Review

The journal operates single blind peer review. Manuscripts sent out for peer review are evaluated by at least one independent reviewer (often two or more). Authors are welcome to suggest independent reviewers to evaluate their manuscript, as well as request individuals or laboratories. All recommendations are considered, but the choice of reviewers is at the editors' discretion. To expedite the review process, only papers that seem most likely to meet editorial criteria are sent for external review. Papers judged by the editors to be of insufficient general interest or otherwise inappropriate are rejected promptly without external review. The editors then make a decision based on the reviewers' evaluations:

- Accept, with or without editorial revisions.
- Revise, with the author addressing concerns raised by the reviewers before a final decision is reached.
- Reject, but indicate to the authors that further work might justify a resubmission.
- Reject outright, typically on grounds of specialist interest, lack of novelty, insufficient conceptual advance or major technical and/or interpretational problems.

Selecting Peer Reviewers

Reviewer selection is critical to the publication process, and the editors' choice is based on many factors, including expertise, reputation, and specific recommendations. A reviewer may decline the invitation to evaluate a manuscript where there is a perceived conflict of interest (financial or otherwise).

Appeals

Even in cases where editors did not invite resubmission, some authors ask the editors to reconsider a rejection decision. These are considered appeals, which, by policy, must take second place to the normal workload. In practice, this means that decisions on appeals often take several weeks. Only one appeal is permitted for each manuscript, and appeals can only take place after peer review.

Decisions are reversed on appeal only if the editors are convinced that the original decision was a serious mistake, not merely a borderline call that could have gone either way. Further consideration may be merited if a referee made substantial errors of fact or showed evidence of bias, but only if a reversal of that referee's opinion would have changed the original decision. Similarly, disputes on factual issues need not be resolved unless they were critical to the outcome. Thus, after careful consideration of the authors' points, most appeals are rejected by the editors.

If an appeal merits further consideration, the editors may send the authors' response or the revised paper to one or more referees, or they may ask one referee to comment on the concerns raised by another referee. On occasion, particularly if the editors feel that additional technical expertise is needed to make a decision, they may obtain advice from an additional referee.

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